AIE1901: AI Exploration I

Lab 2: Exploring Distribution Shift

Core marks: 50

Background

The primary goal of this laboratory is to compare sample sizes and model types, then examine what happens when data collection conditions change. The lab has two parts:

- **Part 1: In-domain evaluation.** Train four models using Condition A images: two sample sizes and two model types. Evaluate them on separate Condition A images and inspect their learning curves.
- **Part 2: Out-of-domain evaluation.** Keep one model from Part 1 fixed. Compare its results on Condition A and Condition B, which differs in one selected collection factor.

In this lab, **in-domain** means that test images follow the same collection conditions as the uploaded images. **Out-of-domain** means that a selected collection condition has changed. Both test sets contain the same classes; out-of-domain does not mean a new class. Such a systematic difference in the data is called **distribution shift**.

Have your Lab 1 image datasets and external test samples ready before class. If your Lab 1 project used audio or poses, download the image dataset provided on Blackboard for this lab instead.

Example: Rock-Paper-Scissors

Suppose a rock-paper-scissors model is trained using images of all three gestures photographed at a close distance, so that the hand occupies a large area of each image. The same gestures may be more difficult to recognise when photographed from farther away, even though their class labels remain unchanged.

To investigate a change in camera-to-hand distance and apparent gesture size, define the conditions as follows:

- **Condition A:** Close-up images in which the hand gesture occupies a large area of the image. Use separate images for the Uploaded Dataset and the In-Domain Test Set.
- **Condition B:** Images taken from farther away, in which the hand gesture occupies a smaller area of the image. Use these images for the Out-of-Domain Test Set.

Both conditions may include varied participants, cameras, backgrounds, lighting, and viewing angles. Make the range and balance of these factors as comparable as possible across the two conditions, so that apparent gesture size is the main systematic difference. The hand must remain visible, and each gesture must still be recognisable.



(a) Condition A: The gesture occupies a large area of the image.



(b) Condition B: The gesture occupies a smaller area of the image.

Figure 1: The gesture label stays the same while its apparent size changes.

In Part 1, compare the four models using only the In-Domain Test Set. In Part 2, use the Standard model trained with 100 images per class to compare both test sets. This examines whether the same model can recognise gestures that appear smaller than those in its Uploaded Dataset.

This is an example only. Choose conditions that fit your own project and available data.

Lab Report and Submission

Work in your Lab 1 groups and follow the academic-integrity requirements in Lab 1. Each group must submit one ZIP file named `Lab2_TeamNumber.zip` to the Lab 2 submission area on Blackboard. Only one member needs to submit on behalf of the group.

At the top of the report, list every group member's name, student ID, and **percentage contribution** to the report. The percentages must sum to 100%. Also briefly describe the tasks or report sections completed by each member.

Include the PDF report, all four downloaded Teachable Machine projects, and both external test sets. Organise the contents as follows:

```
Lab2_TeamNumber/
|-- report.pdf
|-- standard_100.tm
|-- standard_20.tm
|-- embedded_100.tm
|-- embedded_20.tm
|-- testdataset/
    |-- in_domain/
    |   |-- class_1/
    |   |   |-- <test images>
    |   |-- class_2/
    |   |   |-- <test images>
    |   |-- ...
    |   |-- class_N/
    |       |-- <test images>
    |-- out_of_domain/
        |-- class_1/
        |   |-- <test images>
        |-- class_2/
        |   |-- <test images>
        |-- ...
        |-- class_N/
            |-- <test images>
```

The `in_domain` and `out_of_domain` folders contain the Condition A and Condition B test images, respectively. Here, N is the number of classes in your project. Use your actual class names and include every external test image, grouped by its true class. `<test images>` is a placeholder, not a required filename.

Part 1: In-Domain Evaluation (30 marks)

Investigate sample size and model complexity using the larger Standard and smaller Embedded models with the same In-Domain Test Set. Use the training curves and in-domain results to assess underfitting and overfitting.

Step 1: Prepare the Condition A Datasets *(Include at least one table; 8 marks)*

Continue with your Lab 1 image topic and all of its classes, keeping their definitions unchanged. Use Dataset 1, Dataset 2, and the Lab 1 external test dataset as one sample pool. If Lab 1 used audio or poses, use the image dataset provided on Blackboard instead.

Choose one collection factor, such as background, lighting, viewing angle, or camera distance, and define **Condition A** using an observable description. Choose a condition for which other conditions are also available for Part 2. Images must remain recognisable and include every class. Other collection factors may remain varied.

Prepare the following datasets:

- **100-sample dataset:** Select exactly 100 Condition A images per class for uploading.
- **20-sample dataset:** Randomly select 20 images per class from the same 100-sample dataset. Keep its collection conditions as representative of the larger dataset as possible.
- **In-Domain Test Set:** Select at least 10 additional Condition A images per class, with the same number for every class. Keep the range and balance of other collection factors comparable to those in the uploaded dataset.

The 20-sample dataset is a subset of the 100-sample dataset. Neither may contain an In-Domain Test Set image. Avoid putting near-identical frames into both the uploaded and test datasets. Do not upload test images for training. Select samples without referring to any previous prediction results. A sample's role in Lab 1 does not determine its role in Lab 2.

Record the definition of Condition A, sample sources, and the number of images per class in each dataset. If eligible images are insufficient, choose another condition or collect the missing images before training.

Step 2: Train and Evaluate Four Models *(Include settings and results tables, and four Under the hood screenshots; 12 marks)*

Teachable Machine provides two image model options, shown in Figure 2. The **Standard image model** uses 224 × 224 colour images and has an approximate model size of 5 MB. The smaller **Embedded image model** uses 96 × 96 greyscale images and has an approximate model size of 500 KB. The two models differ in model size, input resolution, and colour information.

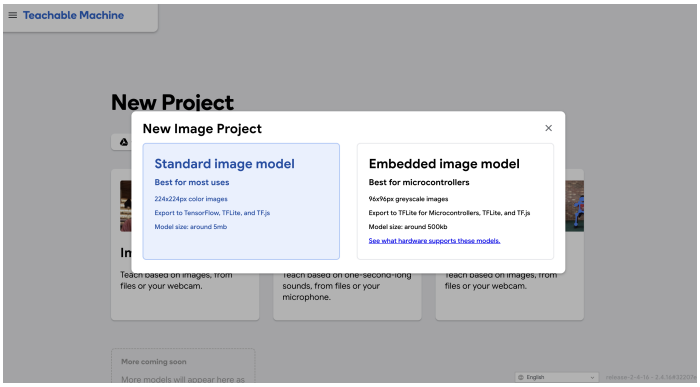


Figure 2: Standard and Embedded image model options in Teachable Machine.

1. **Train the four combinations.** *(4 marks)*

   Train Standard and Embedded models with each of the two datasets from Step 1. For each dataset size, upload exactly the same images to both model types. Keep the class definitions, epochs, batch size, and learning rate the same for all four runs. Record these settings and the uploaded and In-Domain Test Set sample counts for each run. Select **Download project as file** for each project and save the four files using the names listed in the submission requirements. Train Standard with 100 samples per class last and keep its trained project open for Part 2.

2. **Evaluate on the In-Domain Test Set.** *(4 marks)*

   Test every model using every image in the same fixed In-Domain Test Set. Record the number of

correct predictions and overall test accuracy. For Standard with 100 samples per class, also record the correct prediction count and accuracy for each class; these results will be reused in Part 2.

$$\text{Test Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Number of Test Samples}} \times 100\%. \tag{1}$$

Complete Table 1 with each model's final training accuracy and external in-domain accuracy. Convert accuracy values between 0 and 1 into percentages. Do not use the platform's internal test accuracy for the in-domain test column.

3. **Save the learning curves.** *(4 marks)*

   For each model, open **Under the hood** and capture one screenshot showing both **Accuracy per epoch** and **Loss per epoch**, as illustrated in Figure 3. Include four screenshots in total, each labelled with its model type and uploaded sample count per class.

Uploaded samples per class	Model	Training accuracy (%)	In-domain test accuracy (%)
100	Standard	–	–
100	Embedded	–	–
20	Standard	–	–
20	Embedded	–	–

Table 1: Part 1: Compare sample sizes and model types using the same external In-Domain Test Set.

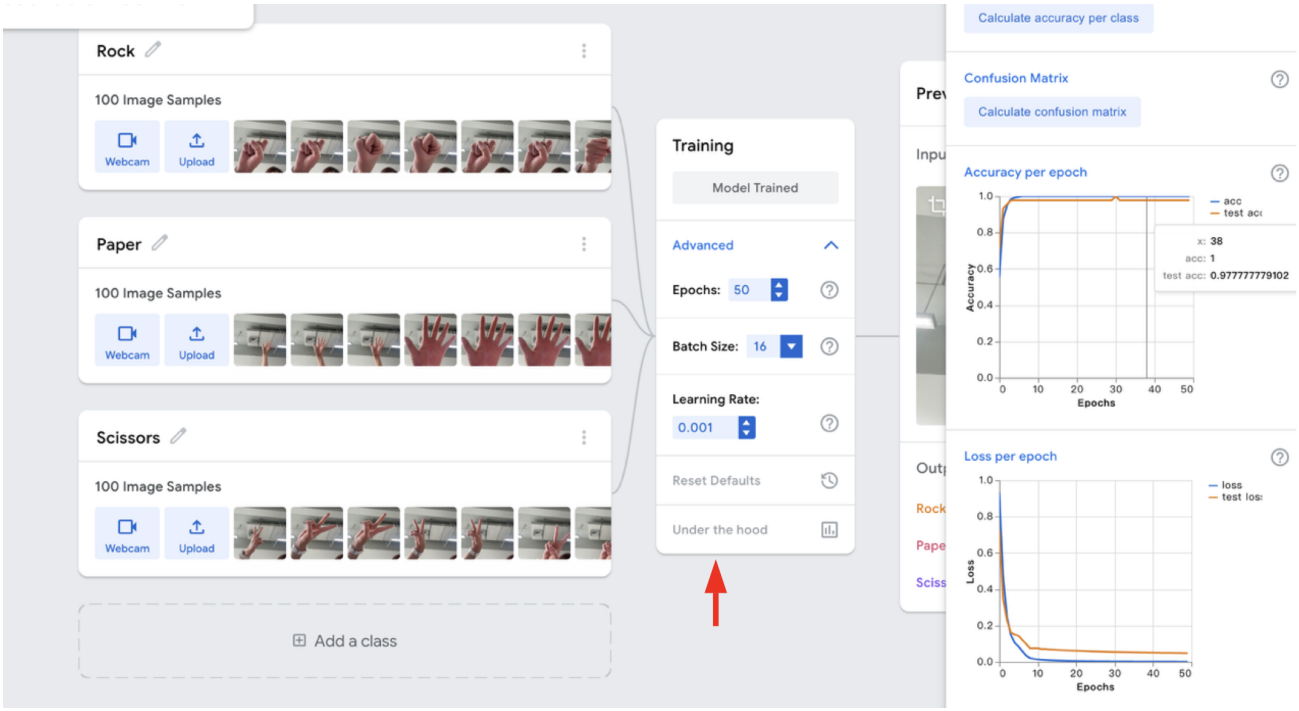


Figure 3: Open Under the hood to display the accuracy and loss curves. Submit one such screenshot for each of the four experiments.

The curves labelled **test acc** and **test loss** use the platform's internal test samples. Analyse these curves separately from the external in-domain test accuracy in Table 1.

Step 3: Compare the In-Domain Results *(Include one plot and brief answers; 10 marks)*

Plot the four in-domain test accuracies in one bar chart. Answer the following questions using only the Part 1 results. Brief answers are sufficient; no extended discussion is required.

1. **Sample size.** *(3 marks, including the plot)* For each model type, did increasing the uploaded samples from 20 to 100 per class make in-domain accuracy higher, lower, or unchanged? Report the two accuracies for each comparison.
2. **Model complexity (Standard versus Embedded).** *(3 marks)* At each sample size, did the larger Standard model have higher, lower, or the same in-domain accuracy as the smaller Embedded model? Support each answer with the recorded accuracies.
3. **Fitting behaviour.** *(4 marks)* For each of the four models, choose the description best supported by your results: **underfitting**, **overfitting**, or **no clear evidence of either**. Give one short reason for each choice, using a feature of the accuracy or loss curves and the external in-domain result.

Typical signs of **underfitting** include poor performance on both the training data and in-domain test data, with training loss remaining high. Signs of **overfitting** include training performance improving while internal test loss rises, or substantially better training than in-domain test performance. These are clues, not automatic rules; a small difference alone is not enough to diagnose overfitting.

Part 2: Out-of-Domain Evaluation (20 marks)

Use only the **Standard model with 100 uploaded samples per class** from Part 1. Keep this same trained model fixed and compare Condition A with Condition B. Do not retrain the model, change its parameters, or upload Condition B images for training. This part examines changed data conditions.

Step 4: Prepare the Out-of-Domain Test Set *(Include at least one table; 6 marks)*

Define **Condition B** by changing the collection factor selected in Step 1. Record the selected factor and the definitions of Conditions A and B. Both conditions must contain every class, and all images must remain recognisable. Keep the range and balance of other factors as comparable as possible; they do not need to be limited to a single participant, background, or lighting condition.

Select Condition B images from your remaining sample pool. Use the same number of images per class as in the In-Domain Test Set, with at least 10 per class. The Out-of-Domain Test Set must not share images with either uploaded dataset or the In-Domain Test Set. Avoid near-duplicate images across the sets. Define the selection rules before testing, and do not refer to any previous prediction results when selecting images. If necessary, collect only the missing Condition B images; keep the Part 1 datasets unchanged.

Record the source and number of out-of-domain images for each class.

Step 5: Evaluate the Same Model on Both Conditions *(Include at least one table; 6 marks)*

Reuse the Standard 100-sample model's in-domain results from Step 2. Test the same trained model on every Out-of-Domain Test Set image. For each condition, record the sample count, correct prediction count, and accuracy for each class and for the whole test set. Use Equation (1) for both test accuracies.

Calculate the change in overall accuracy:

$$\text{Accuracy Change (percentage points)} = A_{\text{out}} - A_{\text{in}}. \tag{2}$$

Here, A_{in} and A_{out} are the overall in-domain and out-of-domain accuracies expressed as percentages. A positive value indicates higher out-of-domain accuracy; a negative value indicates lower out-of-domain accuracy.

Step 6: Compare Conditions A and B *(Include one plot, two sample images, and brief answers; 8 marks)*

1. **Plot the results.** *(2 marks)* Plot the Standard 100-sample model's in-domain and out-of-domain accuracies, showing the overall results and the results for each class.
2. **Show one pair of samples.** *(2 marks)* Select one in-domain and one out-of-domain test image from the same class. Use an incorrectly classified out-of-domain image if one is available; otherwise,

use a correctly classified image. For each image, show its condition, true class, predicted class, and confidence for the predicted class.

3. **Give a short conclusion.** *(2 marks)* Was overall out-of-domain accuracy higher, lower, or unchanged? Report both accuracies and identify the class with the largest accuracy change in either direction. For your sample pair, describe the selected visual difference in one sentence and state whether each image was classified correctly.
4. **Reflection question.** *(2 marks)* Why should Conditions A and B differ mainly in one collection factor? If the viewing angle, lighting, and background all change at the same time, what can and cannot be concluded from a difference in accuracy?

AI Use Declaration

(Required; not graded)

Did any member of your group use additional AI tools (e.g., ChatGPT), other than Teachable Machine, while completing this lab? If yes, name each tool and explain what you used it for. If not, state that no additional AI tools were used.